

Literature-Implied Answer to Conjecture 3.6: Semialgebraic Hypographs Have Finitely Many Corners

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Status: full resolution implied by a classical theorem. If the hypograph is semialgebraic, then its upper fiber endpoint f is a semialgebraic function. Semialgebraic C^1 cell decomposition makes f C^1 away from finitely many points. Thus the conjectured statement is true.

1. Source conjecture

Paulsen and Solazzo pose the following as Conjecture 3.6 on arXiv PDF page 26 [1].

Conjecture 3.6 (Paulsen–Solazzo). *Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous and nonnegative, and set*

$$C_f = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq f(x)\}.$$

If f is nondifferentiable at infinitely many points, then C_f is not semialgebraic.

The source says this was the only obstruction to its proposed nonsemialgebraic operator-algebra ball.

2. Known regularity theorem

The C^1 cell decomposition theorem for definable functions says that if $X \subseteq \mathbb{R}^n$ and $g : X \rightarrow \mathbb{R}$ are definable, then there is a finite decomposition of $\mathbb{R}^n$ into C^1 cells partitioning X such that g restricts to a C^1 function on every cell contained in X . See Bhardwaj–van den Dries, Theorem A.14 (PDF page 35) [2]. Semialgebraic sets and functions are definable in the real field, so the theorem applies to them.

In dimension one, a finite cell decomposition consists of finitely many points and finitely many open intervals. Consequently, every semialgebraic function on $[0, 1]$ is C^1 off a finite set.

3. Deduction

Proposition. *Conjecture 3.6 is true.*

Proof. Suppose toward a contradiction that C_f is semialgebraic. Its upper-fiber endpoint set is

$$\Gamma = \{(x, y) \in C_f : \neg \exists t \in \mathbb{R} (t > y \wedge (x, t) \in C_f)\}.$$

Semialgebraic sets are closed under Boolean operations and coordinate projections (Tarski–Seidenberg), so Γ is semialgebraic. For every $x \in [0, 1]$, the vertical fiber of C_f is exactly $[0, f(x)]$; hence

$$\Gamma = \{(x, f(x)) : 0 \leq x \leq 1\}.$$

Thus f is a semialgebraic function.

Apply C^1 cell decomposition to $f : [0, 1] \rightarrow \mathbb{R}$. There is a finite partition of $[0, 1]$ into point cells and interval cells such that f is C^1 on each interval cell. Therefore every possible point of nondifferentiability lies among the finitely many point cells and interval endpoints. This contradicts the hypothesis that f is nondifferentiable at infinitely many points. Hence C_f is not semialgebraic. $\square$

4. Scope and provenance

The proof resolves the source conjecture completely, including the case where f vanishes on a nontrivial set: the maximal-fiber formula defining Γ remains valid. By the source’s Proposition 3.7, its stated operator-algebra consequence is therefore unconditional. This note does not independently audit the $\mathcal{O}_3/\mathcal{O}_4$ dimension inconsistency in the wording and proof surrounding Conjecture 3.5 and Proposition 3.7.

This is not claimed as a new theorem. It is a direct identification of Conjecture 3.6 as a consequence of classical semialgebraic regularity. A bounded exact-phrase, author/title, citation, local-corpus, and web search through 22 July 2026 found no later paper explicitly recording this particular implication.

References

- [1] V. I. Paulsen and J. P. Solazzo, *Interpolation and Balls in $\mathbb{C}^k$* , J. Operator Theory **60** (2008), 379–398; arXiv:math/0604497.
- [2] N. Bhardwaj and L. van den Dries, *On the Pila–Wilkie theorem*, arXiv:2010.14046, Appendix A, Theorem A.14.